Tomato Diseases

Identification, causes, favourable conditions, management and verified remedies

Twelve major diseases with a Maharashtra focus and all-India context

Primary sources: TNAU Agritech Portal, CIB&RC (Insecticides Act, 1968) Major Uses of Pesticides, ICAR / ICAR-IIHR publications, MPKV Rahuri, UC IPM and other university extension material. Compiled 18 October 2026.

Contents

About this document

Scope and method

Each disease chapter follows the same six headings: identification; causes and progression; favourable conditions; prevention and management; remedies and treatment; and sources. Disease descriptions come from TNAU, ICAR and university extension pages. Chemical recommendations are included only where they could be matched to the CIB&RC compilation of registered uses on tomato or to a published TNAU / ICAR recommendation, and each is tagged to show how it was verified.

Verification tags used in the chemical tables

V1 - Verified against CIB&RC. The product is listed for TOMATO and this disease/pest in the CIB&RC 'Major Uses of Pesticides' compilation - fungicides up to 30/09/2025 (CIB); insecticides from the 2019/2021 versions (CIBI). Doses are reproduced as printed.

V2 - University recommendation only. Published by TNAU (Agritech portal / Crop Production Guide - Horticulture 2020). The tomato label claim could NOT be matched in the CIB&RC lists reviewed, so confirm the current registration and product label before use.

V3 - ICAR publication. Dose taken from an ICAR (Indian Horticulture) article; same confirmation step as V2.

-- - Not recommended / no chemical option verified. The sources reviewed do not support a chemical recommendation, so none is given.

Important cautions

- The fungicide entries tagged V1 come from the CIB&RC compilation up to 30/09/2025, read from a copy on a university website, not the live register. The insecticide entries come from older CIB&RC versions (2019 / 2021). Registrations, label claims and permitted uses change; some listed active ingredients may since have been restricted or withdrawn. Always check the current label and the live register (cibrc.gov.in) and follow the pre-harvest interval (PHI, in days) printed on the label.
- CIB&RC doses are 'per hectare (ha)' unless stated; TNAU doses are 'per acre (ac)'. 1 ha = 2.47 ac. Water volumes depend on crop stage and sprayer type. In a few combination entries the unit printed beside a dose is unclear; those rows say 'as printed'.
- Rotate fungicide groups across sprays to slow resistance. Use PPE, follow label safety instructions and keep products away from water bodies.
- No MPKV Rahuri or PDKV Akola spray schedule for tomato diseases was retrieved during this review, so none is included. Maharashtra-specific notes are limited to what the cited sources state.
- Cross-check finding on vector insecticides: several TNAU per-acre doses for leaf curl (for example thiamethoxam 25 WG 200 g/ac and imidacloprid 17.8 SL 200 ml/ac) are numerically about the same as the CIB&RC per-hectare doses (thiamethoxam 25 WG 200 g/ha; imidacloprid about 150-175 ml/ha), that is, roughly 2.5 times higher when converted. Use the CIB&RC / label dose, not the TNAU per-acre figure, until TNAU clarifies.

Summary table

Disease	Pathogen / type	Main non-chemical measures	Chemical status
Damping Off	Pythium aphanidermatum (also Rhizoctonia solani reported) (Fungal / oomycete (seed- and soil-borne))	Raised beds, drainage, thin sowing; Trichoderma / Bacillus seed treatment	Nursery/seed treatment: V1 (captan, metalaxyl-M, fosetyl-Al) and V2 options
Early Blight	Alternaria solani (Fungal (seed-, debris-borne))	Debris removal, rotation, resistant hybrids (Arka Rakshak, Samrat, Abhed)	Many V1 options (singles and combinations); rotate groups, check PHI
Late Blight	Phytophthora infestans (Oomycete (fungus-like), airborne)	Arka Abhed, drainage, mulch, preventive copper sprays	Many V1 options incl. cyazofamid, mandipropamid, combinations; early preventive action
Septoria Leaf Spot	Septoria lycopersici (Fungal (debris-borne))	Debris removal, rotation, reduce leaf wetness	V1 options with a Septoria claim (combinations); general leaf-spot V1
Fusarium Wilt	Fusarium oxysporum f. sp. lycopersici (Fungal (soil-borne, vascular))	Resistant varieties, rotation, solarization, clean seedlings	No curative chemical; V2 seed treatment
Bacterial Wilt	Ralstonia solanacearum (Bacterial (soil- and water-borne))	Resistant hybrids, rotation, restricted irrigation	No effective bactericide; V2 soil drench only
Bacterial Stem and Fruit Canker	Clavibacter michiganensis subsp. michiganensis (Bacterial (seed- and debris-borne))	Clean seed, sanitation, rotation, disinfect tools	None verified (tag --)
Tomato Leaf Curl Disease	Tomato leaf curl virus (Begomovirus); vector: whitefly Bemisia tabaci (Viral (whitefly-transmitted))	Protected nursery, barrier crop, sticky traps, mulch, resistant hybrids	Vector insecticides: V1 (older CIB&RC lists) and V2; TNAU per-acre doses look too high
Tomato Spotted Wilt / Bud Necrosis	Tomato spotted wilt virus / Groundnut bud necrosis virus (Orthotospovirus); vector: thrips (Viral (thrips-transmitted))	Healthy seedlings, roguing to 45 days, barrier crop	Vector insecticides: V1 (cyantraniliprole, imidacloprid WG) and V2
Bacterial Leaf Spot (Bacterial Spot)	Xanthomonas axonopodis pv. vesicatoria (Bacterial (seed- and debris-borne; rain-splashed))	Disease-free seed; sanitation; limit leaf wetness	V1: streptocycline (40-100 ppm), thiophanate-methyl + kasugamycin
Leaf Mould	Passalora fulva (Cladosporium fulvum); Pseudocercospora fuligena also reported (Fungal (debris-, seed- and wind-borne; mainly protected cultivation))	Airflow, drip irrigation, residue removal, resistant varieties	One V1 claim (zineb, grey leaf mould)
Tomato Mosaic Virus (and related mosaic viruses)	Tomato mosaic virus (ToMV) and tomato mottle mosaic virus (ToMMV) - tobamoviruses; cucumber mosaic virus (CMV) is also blamed for mosaic in Maharashtra (Viral (seed-, contact- and sap-borne; CMV aphid-borne))	Clean seed, nursery biosafety, sanitation, weed removal	None (no chemical cure; tag --)

Quick separation of look-alikes: small angular water-soaked spots with a yellow halo point to bacterial spot; yellow patches above and velvety olive mould below point to leaf mould; mottled mosaic or shoestring leaves point to mosaic virus; concentric rings on lower leaves point to early blight; water-soaked blight with white growth in cool, wet weather points to late blight; many small grey-centred spots point to Septoria; sudden wilt with white ooze points to bacterial wilt; yellowing with brown vascular tissue and no ooze points to Fusarium wilt; leaf rolling and stunting points to leaf curl, whereas ring spots and bronzing point to spotted wilt / bud necrosis.

1. Damping Off

Tomato | Pythium aphanidermatum (also Rhizoctonia solani reported) | Fungal / oomycete (seed- and soil-borne)

At a glance. Seedling disease of the nursery. Seedlings rot at the soil line and topple over. Worst in wet, cloudy, cool spells with crowded beds.

1. Identification

Symptoms

- Crop / stage: tomato seedlings in nursery beds (pre- and post-emergence).
- Pre-emergence phase: complete rotting of seedlings before they emerge, leaving patchy, gappy beds.
- Post-emergence phase: infected tissue at the base becomes soft and water-soaked; seedlings topple over or collapse.

Distinguishing features

- Collapse happens at the soil line while the upper seedling is still green, and spreads in patches within the bed.
- Wilt diseases (Fusarium, bacterial wilt) affect transplanted/older plants and show vascular browning; damping off is a nursery-stage problem.

2. Causes and progression

- Pathogen: Pythium aphanidermatum (Edson) Fitzp. - non-septate, hyaline mycelium; bi-flagellate zoospores (asexual); thick-walled oospores as resting spores (T1).
- Rhizoctonia solani is also listed as a damping-off pathogen of tomato (COL).
- Primary survival: soil, seed and water. Secondary spread: infected debris (T1).
- Progression: oospores/mycelium in soil infect seed and young roots/stem base when soil is wet; infected tissue softens, seedlings collapse and the pathogen spreads to neighbouring seedlings through water and debris.

3. Favourable conditions

- High humidity and high soil moisture; cloudy weather (T1).
- Low temperature, below 24 C for a few days (T1).
- Crowded seedlings, dampness from heavy rain, poor drainage and excess soil solutes (T1).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Avoid water stagnation; use raised nursery beds of about 15 cm height (T1).
- Provide better drainage, sow thinly and burn trash on the bed surface before sowing (COL).

Maharashtra note

No Maharashtra-specific recommendation was found for this disease in the sources reviewed.

5. Remedies and treatment

Biological / soil-applied options

- Seed treatment 24 h before sowing: Trichoderma asperellum 4 g/kg seed OR Bacillus subtilis 10 g/kg seed (T1).
- Soil application of Bacillus subtilis 2.5 kg/ha mixed with 50 kg FYM (T1).
- CIB&RC lists Trichoderma viride 1% WP as registered on tomato for seedling wilt (Fusarium oxysporum) and damping off (Pythium aphanidermatum, Rhizoctonia solani); apply as per the product label (CIB).

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Captan 75% WP	Damping off in nursery (soil drench)	0.25% (2.5 kg in 1000 L water per ha)	1000 L	V1	CIB&RC 2025; PHI NA
Captan 75% WS	Damping off, seed treatment	Registered on tomato; the printed per-kg seed figure is ambiguous, so	-	V1	CIB&RC 2025; PHI not stated

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
		follow the product label			
Metalaxyl-M 31.8% ES	Damping off (seed dresser)	2.0 ml per kg seed (a.i. 0.64 g) as printed; TNAU: 2 ml/kg seed	-	V1	CIB&RC 2025; PHI seed dresser
Fosetyl-Al 80% WP	Tomato seedlings damping off	2.5-3.0 g formulation per litre, applied at 1 L per sq m (as printed)	1 L/sq m	V1	CIB&RC 2025; PHI not stated
Copper oxychloride 50% WP	Nursery soil drench	2.5 g/litre; drench 4 litres per sq. m	-	V2	TNAU (T1)

Seed treatment and nursery drench are the points where chemicals are used; foliar sprays are not part of damping-off management. Metalaxyl-M seed dressing (TNAU V2 earlier) and fosetyl-Al are now matched to the CIB&RC 2025 list.

6. Sources and citations

[T1] TNAU Agritech Portal - Tomato damping off (Pythium aphanidermatum). Source cited by TNAU: Crop Production Guide - Horticulture 2020. https://agritech.tnau.ac.in/crop_protection/tomato_diseases_1.html

[COL] Open textbook 'Integrated Disease Management', Commonwealth of Learning - Ch. 3: Tomato diseases and their management. https://openbooks.col.org/integrateddiseasemanagement/chapter_3-tomato-diseases-and-their-management.html

[T0] TNAU Agritech Portal - Tomato: crop protection (disease index). https://agritech.tnau.ac.in/crop_protection/crop_prot_crop%20diseases_veg_tomato.html (updated Jun 2023)

[CIB] Central Insecticide Board & Registration Committee (CIB&RC), DPPQS, Govt. of India: 'Major Uses of Pesticides (registered under the Insecticides Act, 1968) - Fungicides', up to 30/09/2025 (single products pp. 2-41; combinations pp. 42-79). Copy reviewed (hosted on a university website, not the live register): https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Official portal: http://cibrc.nic.in/ . The compilation states it is for guidance and not for legal purposes.

[CIB12] CIB&RC 'Major Uses of Pesticides - Fungicides' as on 30.09.2012 (older compilation, used for the first version of this document). https://agritech.tnau.ac.in/crop_protection/pdf/6_Major_use_fungicides.pdf

2. Early Blight

Tomato | Alternaria solani | Fungal (seed-, debris-borne)

At a glance. Brown target-board spots on older leaves, stem collar and fruit. Favoured by warm, humid, rainy weather. Second most serious tomato disease after leaf curl in an ICAR assessment.

1. Identification

Symptoms

- Brown spots with concentric rings in a 'bull's eye' pattern, surrounded by a yellow margin (T2).
- First appears as small dark lesions on the older (lower) foliage; the fungus can attack at any growth stage (Vikaspedia/NCIPM; UM).
- Fruit is infected through the calyx or stem attachment, giving brown concentric rings on the fruit (T2).

Distinguishing features

- Late blight lesions are light tan with a pale-green halo and occur on young leaves as well; early blight is found mainly on the lower leaves (NUN).
- Gray leaf spot lesions are smaller and lighter brown, and the centre tends to crack (NUN).
- Bacterial spot lesions have no concentric zones or prominent halo (UM).

2. Causes and progression

- Pathogen: Alternaria solani - coloured, septate mycelium; coloured beaked conidia with transverse and longitudinal septa, borne singly or in chains (T2).
- Survival: infected plant debris. Primary spread: infected seed. Secondary spread: conidia carried by rain and wind (T2).
- Progression: conidia germinate on wet leaves, lesions expand with concentric rings on lower leaves, spores are splashed/blown to upper foliage and fruit; severe cases cause leaf drop and fruit rot at the calyx.

3. Favourable conditions

- Warm, humid weather of 24-29 C with rainy and wet conditions (T2).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Remove and destroy crop debris after harvest (T2).
- Crop rotation away from tomato (T2).
- Seed treatment with Trichoderma asperellum 4 g/kg seed (T2).
- Resistant hybrids: ICAR-IIHR Arka Rakshak and Arka Samrat are reported resistant to early blight along with leaf curl and bacterial wilt (AR, IH88); Arka Abhed (H-397) carries early blight and late blight resistance (AA). An ICAR article names resistant varieties as the single best approach to early blight (IH82).

Maharashtra note

NCIPM-based list records early blight on tomato in Maharashtra (NCI). MPKV Rahuri's Tomato Improvement Scheme lists early blight resistance as a breeding objective (MP).

5. Remedies and treatment

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Azoxystrobin 23% SC	Early & late blight	500 ml/ha (TNAU: 200 ml/ac)	500 L	V1	CIB&RC 2025; PHI 3 d
Pyraclostrobin 20% WG	Early blight	375-500 g/ha (TNAU: 200 g/ac)	500 L	V1	CIB&RC 2025; PHI 3 d
Mancozeb 75% WG	Early blight	1000 g/ha	500 L	V1	CIB&RC 2025; PHI 5-6 d
Mancozeb 35% SC	Early & late blight	0.5% formulation (TNAU: 1 kg/ac)	500 L	V1	CIB&RC 2025; PHI 10 d
Metiram 70% WG	Alternaria blight	2500 g/ha (TNAU: 1.0 kg/ac)	500-750 L	V1	CIB&RC 2025; PHI 6 d
Metiram 55% + Pyraclostrobin 5% WG	Early blight	1500-1750 g/ha (TNAU: 600 g/ac)	500 L	V1	CIB&RC 2025; PHI 5 d

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Iprodione 50% WP	Early blight	1.5 kg/ha (TNAU: 600 g/ac)	500 L	V1	CIB&RC 2025; PHI 15 d
Captan 75% WP	Early & late blight	1667 g/ha	1000 L	V1	CIB&RC 2025; PHI 6 d
Captan 50% WP	Early & late blight	2.5 kg/ha	750-1000 L	V1	CIB&RC 2025; PHI not stated
Copper oxychloride 50% WP	Early blight, late blight, leaf spot	2.5 kg/ha	750-1000 L	V1	CIB&RC 2025; PHI not stated
Copper hydroxide 61.41% WG	Early & late blight	1250 g/ha	500 L	V1	CIB&RC 2025; PHI 3 d
Copper sulphate 2.62% SC	Early & late blight	1 L/ha	500 L	V1	CIB&RC 2025; PHI 3 d
Zineb 75% WP	Early blight, late blight, grey leaf mould	1.5-2.0 kg/ha (TNAU: 800 g/ac)	750-1000 L	V1	CIB&RC 2025; PHI not stated
Ziram 80% WP	Early blight	1.5-2.0 kg/ha	750-1000 L	V1	CIB&RC 2025; PHI 3 d
Kasugamycin 3% SL	Early blight	1000-1250 ml/ha	500-600 L	V1	CIB&RC 2025; PHI 3 d
Iprobenfos (Kitazin) 48% EC	Early blight	0.2% formulation (200 ml/100 L)	as required	V1	CIB&RC 2025; PHI 5 d
Picarbutrazox 9.53% SC	Early & late blight	1000-1250 ml/ha	500 L	V1	CIB&RC 2025; PHI 7 d
Thifluzamide 24% SC	Early blight	500 ml/ha	500 L	V1	CIB&RC 2025; PHI 7 d
Tebuconazole 38.39% w/w SC	Early blight and anthracnose	625 ml/ha	500 L	V1	CIB&RC 2025; PHI 5 d
Famoxadone 16.6% + Cymoxanil 22.1% SC	Early & late blight	500 (g or ml)/ha	500 L	V1	CIB&RC 2025; PHI 3 d
Azoxystrobin 18.2% + Difenoconazole 11.4% SC	Early & late blight	1 ml/L (0.1%); TNAU: 200 ml/ac	500 L	V1	CIB&RC 2025; PHI 5 d
Tebuconazole 50% + Trifloxystrobin 25% WG	Early blight	350 g/ha (TNAU: 140 g/ac)	500 L	V1	CIB&RC 2025; PHI 5 d
Azoxystrobin 8.3% + Mancozeb 66.7% WG	Early & late blight	1500 g/ha	500 L	V1	CIB&RC 2025; PHI 5 d
Azoxystrobin 11.5% + Mancozeb 30% WP	Early & late blight	750-875 g/ha	500 L	V1	CIB&RC 2025; PHI 3 d
Mancozeb 40% + Azoxystrobin 7% OS	Early & late blight	1500 g/ha	500 L	V1	CIB&RC 2025; PHI 5 d
Azoxystrobin 11% + Tebuconazole 18.3% SC	Early blight	750 g or ml/ha (as printed)	500 L	V1	CIB&RC 2025; PHI 7 d
Azoxystrobin 25% + Boscalid 35% WG	Early blight and Septoria leaf spot	500 g/ha	500 L	V1	CIB&RC 2025; PHI 10 d
Boscalid 25.2% + Pyraclostrobin 12.8% WG	Early blight	500 g/ha	500 L	V1	CIB&RC 2025; PHI 10 d
Captan 70% + Hexaconazole 5% WP	Early blight, powdery mildew, leaf spot, fruit rot	750 g/ha	500 L	V1	CIB&RC 2025; PHI 7 d
Chlorothalonil 40% + Difenoconazole 4% SC	Early & late blight	1000 ml/ha	500 L	V1	CIB&RC 2025; PHI 7 d
Dimethomorph 12% + Pyraclostrobin 6.7% WG	Early blight	1250 g/ha	500 L	V1	CIB&RC 2025; PHI 10 d
Flubendiamide 7.5% + Kresoxim-methyl 37.5% SC	Early & late blight	667 ml/ha	500 L	V1	CIB&RC 2025; PHI 7 d
Fluxapyroxad 75 g/l + Difenoconazole 50 g/l SC	Early blight and Septoria leaf spot	800-1000 ml/ha	500 L	V1	CIB&RC 2025; PHI 10 d
Fluxapyroxad 250 g/l + Pyraclostrobin 250 g/l SC	Early blight and Septoria leaf spot	200-250 ml/ha	500 L	V1	CIB&RC 2025; PHI 10 d
Kresoxim-methyl 18% + Mancozeb 54% WP	Early blight, late blight, fruit rot	1388 g/ha	500 L	V1	CIB&RC 2025; PHI 7 d
Mefentrifluconazole 200	Early blight and	400-500 ml/ha	500 L	V1	CIB&RC 2025; PHI 7 d

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
g/l + Pyraclostrobin 200 g/l SC	Septoria leaf spot				
Metalaxyl-M 3.3% + Chlorothalonil 33.1% SC	Early & late blight	1000 ml/ha (0.2%)	500 L	V1	CIB&RC 2025; PHI 5 d
Pydiflumetofen 6.89% + Difenoconazole 11.49% SC	Early blight and powdery mildew	500 ml/ha	500 L	V1	CIB&RC 2025; PHI 5 d
Pyraclostrobin 10% + Metiram 30% + Difenoconazole 10% WG	Early & late blight	562.5 g/ha	500 L	V1	CIB&RC 2025; PHI 10 d
Trifloxystrobin 10% + Difenoconazole 12.5% + Sulphur 3% SC	Early & late blight	450 ml/ha	500 L	V1	CIB&RC 2025; PHI 14 d
Trifloxystrobin 25% + Hexaconazole 25% WG	Early blight, late blight, powdery mildew, fruit spot	200 g/ha	500 L	V1	CIB&RC 2025; PHI 7 d
Valifenalate 6% + Mancozeb 60% WG	Early & late blight	2000-2500 g/ha	500 L	V1	CIB&RC 2025; PHI 5 d
Iprovalicarb 5.5% + Propineb 61.25% WP	Early & late blight	2.25-2.5 kg/ha	500 L	V1	CIB&RC 2025; PHI 5 d
Oxathiapiprolin 1.4% + Azoxystrobin 14.46% SC	Late blight, early blight, powdery mildew	800-900 ml/ha	500 L	V1	CIB&RC 2025; PHI 5 d

Follow the label for timing, interval and PHI, and rotate between chemical groups; the long list reflects many CIB&RC tomato entries, so choose by PHI, resistance management and cost. TNAU per-acre values for azoxystrobin, pyraclostrobin, mancozeb 35% SC, metiram, iprodione, zineb, metiram + pyraclostrobin, azoxystrobin + difenoconazole and tebuconazole + trifloxystrobin agree with the CIB&RC per-hectare values. Items earlier tagged V2 (azoxystrobin + difenoconazole; tebuconazole + trifloxystrobin) are now V1.

6. Sources and citations

[T2] TNAU Agritech Portal - Tomato early blight (Alternaria solani). https://agritech.tnau.ac.in/crop_protection/tomato_diseases_2.html

[AA] ICAR-IIHR: Multiple disease resistant tomato hybrid Arka Abhed (H-397). https://iihr.res.in/node/23884

[AR] Lok Sabha / Parliament reply on ICAR-IIHR hybrid 'Arka Rakshak' (resistance to leaf curl, bacterial wilt, early blight). https://eparlib.nic.in/bitstream/123456789/663314/1/17708.pdf

[IH88] Hanson P. et al. Having a basket full of disease-free tomatoes (Arka Samrat, Arka Rakshak). Indian Horticulture, ICAR. https://epubs.icar.org.in/index.php/IndHort/article/view/88525

[IH82] Singh, R.K. & Singh, S. (2019). Managing early blight disease: hindrance to tomato fruit production. Indian Horticulture 61(2). https://epubs.icar.org.in/index.php/IndHort/article/view/88382

[NUN] Nunhems India (company extension page): Tomato disease management - look-alike diseases, begomovirus notes (supplementary, non-university source). https://www.nunhems.com/in/en/disease-management/tomato-disease-management

[UM] UMass Extension: Tomato disease scouting guide for IPM field walk (Septoria, early blight look-alikes). https://ag.umass.edu/sites/ag.umass.edu/files/pdf-doc-ppt/tomato_disease_accessible.pdf

[NCI] NCIPM-based list of tomato pests/diseases and reported states (via Vikaspedia): Tomato pests. https://vikaspedia.in/agriculture/crop-production/integrated-pest-managment/ipm-for-vegetables/ipm-strategies-for-tomato/tomato-pests

[MP] Mahatma Phule Krishi Vidyapeeth, Rahuri - Tomato Improvement Scheme (objectives, released varieties incl. Phule Raja). https://mpkv.ac.in/Uploads/Research/20260819_123737_3fbf99aa75c9858af9fe7ad9fbe7faaf.pdf

[CIB] Central Insecticide Board & Registration Committee (CIB&RC), DPPQS, Govt. of India: 'Major Uses of Pesticides (registered under the Insecticides Act, 1968) - Fungicides', up to 30/09/2025 (single products pp. 2-41; combinations pp. 42-79). Copy reviewed (hosted on a university website, not the live register): https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Official portal: http://cibrc.nic.in/ . The compilation states it is for guidance and not for legal purposes.

[CIB12] CIB&RC 'Major Uses of Pesticides - Fungicides' as on 30.09.2012 (older compilation, used for the first version of this document). https://agritech.tnau.ac.in/crop_protection/pdf/6_Major_use_fungicides.pdf

3. Late Blight

Tomato | Phytophthora infestans | Oomycete (fungus-like), airborne

At a glance. Fast-spreading blight of leaves, stems and fruit in cool, wet, foggy weather. Can destroy a field in days. Preventive sprays and resistant hybrids give the best protection.

1. Identification

Symptoms

- Leaves: water-soaked, black/dark lesions that expand rapidly until the whole leaf is necrotic; white sporulation (sporangia) on the lesion (T8).
- Stems and petioles: dark, greasy-looking patches/brown streaks that spread (T8, IH26).
- Fruit: dark brown, olive-brown greasy lesions leading to soft rot and disintegration (T8, IH26).

Distinguishing features

- Late blight lesions are light tan with a pale-green halo and occur all over the plant including young leaves; early blight sits mainly on lower leaves and shows concentric rings (NUN).
- Late blight fruit lesions are rough and sunken; buckeye rot lesions (soil-contact fruit) are less sharply defined (UGA).

2. Causes and progression

- Pathogen: Phytophthora infestans (Mont.) de Bary - hyaline, coenocytic mycelium; sporangia release bi-flagellate zoospores; thick-walled oospores form sexually (T8).
- Survival and primary spread: infected debris. Secondary spread: wind-borne sporangia (T8).
- Progression: air-borne spores land on wet foliage and infect; infection can spread widely in 4-5 days (IH26).

3. Favourable conditions

- Temperature 15-18 C, relative humidity above 90 %, continuous wet conditions, wind to disperse spores (IH26).
- TNAU: cool nights, warm days and extended wet conditions from rain and fog; optimum temperature for sporulation 18-22 C (T8).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Grow a resistant hybrid: ICAR-IIHR Arka Abhed (H-397) has Ph-2 + Ph-3 late-blight resistance; in two years of natural epidemics the average yield loss was 12.84 % vs 75.5 % (Arka Rakshak) and 79.5 % (NS 501) (RED, AA). Arka Rakshak is resistant to leaf curl and bacterial wilt but susceptible to late blight (RED).
- Proper drainage, crop rotation, field hygiene (remove infected plant parts), adequate spacing and staking for airflow (T8, IH26).
- Black polythene mulch (30 micron) to reduce soil moisture and spread; avoid overhead irrigation; plant windbreaks; avoid late sowing in the monsoon (IH26).
- Soil application of Trichoderma asperellum (viride) 1.0 kg/ac (T8).

Maharashtra note

In the (older) NCIPM-based list the states recorded for tomato late blight do not include Maharashtra (NCI); absence from an old list does not mean absence in the field, so risk depends on local cool, wet, foggy spells; follow local KVK advisories.

5. Remedies and treatment

Biological / soil-applied options

- Soil application of Trichoderma asperellum 1.0 kg/ac (T8).

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Azoxystrobin 23% SC	Early & late blight	500 ml/ha (TNAU: 200 ml/ac)	500 L	V1	CIB&RC 2025; PHI 3 d
Mancozeb 35% SC	Early & late blight	0.5% formulation (TNAU: 1000 g/ac)	500 L	V1	CIB&RC 2025; PHI 10 d

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Mancozeb 75% WP	Late blight	1.5-2.0 kg/ha	750 L	V1	CIB&RC 2025; PHI not stated
Cymoxanil 8% + Mancozeb 64% WP	Late blight	1500 g/ha	500-750 L	V1	CIB&RC 2025; PHI 10 d
Famoxadone 16.6% + Cymoxanil 22.1% SC	Early & late blight	500 (g or ml)/ha	500 L	V1	CIB&RC 2025; PHI 3 d
Captan 75% WP	Early & late blight	1667 g/ha	1000 L	V1	CIB&RC 2025; PHI 6 d
Captan 50% WP	Early & late blight	2.5 kg/ha	750-1000 L	V1	CIB&RC 2025; PHI not stated
Zineb 75% WP	Late blight	1.5-2.0 kg/ha (TNAU: 800 g/ac)	750-1000 L	V1	CIB&RC 2025; PHI not stated
Copper oxychloride 50% WP	Late blight (preventive)	2.5 g/litre, 6 sprays at 6-day intervals plus mulching (ICAR trial); CIB&RC: 2.5 kg/ha in 750-1000 L	750-1000 L	V1	CIB&RC 2025 + ICAR (IH26)
Copper hydroxide 61.41% WG	Early & late blight	1250 g/ha	500 L	V1	CIB&RC 2025; PHI 3 d
Copper sulphate 2.62% SC	Early & late blight	1 L/ha	500 L	V1	CIB&RC 2025; PHI 3 d
Cyazofamid 34.5% SC	Late blight	200 ml/ha (80 g a.i.; TNAU: 80 ml/ac)	500 L	V1	CIB&RC 2025; PHI 3-5 d
Mandipropamid 23.4% SC	Late blight	0.8 ml/L (0.08%)	500 L	V1	CIB&RC 2025; PHI 5 d
Picarbutrazox 9.53% SC	Early & late blight	1000-1250 ml/ha	500 L	V1	CIB&RC 2025; PHI 7 d
Azoxystrobin 18.2% + Difenoconazole 11.4% SC	Early & late blight	1 ml/L (0.1%); TNAU: 200 ml/ac	500 L	V1	CIB&RC 2025; PHI 5 d
Ametoctradin 27% + Dimethomorph 20.27% w/w SC	Late blight	800-1000 ml/ha	500 L	V1	CIB&RC 2025; PHI 3 d
Azoxystrobin 8.3% + Mancozeb 66.7% WG	Early & late blight	1500 g/ha	500 L	V1	CIB&RC 2025; PHI 5 d
Azoxystrobin 11.5% + Mancozeb 30% WP	Early & late blight	750-875 g/ha	500 L	V1	CIB&RC 2025; PHI 3 d
Mancozeb 40% + Azoxystrobin 7% OS	Early & late blight	1500 g/ha	500 L	V1	CIB&RC 2025; PHI 5 d
Mancozeb 69.7% + Cyazofamid 5.3% WG	Late blight and Alternaria leaf spot	1500 g/ha	500 L	V1	CIB&RC 2025; PHI 7 d
Mancozeb 64% + Thiophanate-methyl 12% WP	Late blight, leaf spot, buckeye rot, ring rot	1500 g/ha	750 L	V1	CIB&RC 2025; PHI 7 d
Oxathiapiprolin 0.6% + Mancozeb 60% WG	Late blight	2500 g/ha	500 L	V1	CIB&RC 2025; PHI 5 d
Oxathiapiprolin 1.4% + Azoxystrobin 14.46% SC	Late blight, early blight, powdery mildew	800-900 ml/ha	500 L	V1	CIB&RC 2025; PHI 5 d
Metiram 44% + Dimethomorph 9% WG	Late blight	2500 g/ha	500 L	V1	CIB&RC 2025; PHI 10 d
Fluopicolide 6.25% + Propamocarb 62.5% SC	Late blight	1.5 L/ha	500 L	V1	CIB&RC 2025; PHI 5 d
Cyazofamid 25% + Pyraclostrobin 18% SC	Late blight and early blight	300-350 ml/ha	500 L	V1	CIB&RC 2025; PHI 26 d
Metalaxyl-M 3.3% + Chlorothalonil 33.1% SC	Early & late blight	1000 ml/ha (0.2%)	500 L	V1	CIB&RC 2025; PHI 5 d
Chlorothalonil 40% + Difenoconazole 4% SC	Early & late blight	1000 ml/ha	500 L	V1	CIB&RC 2025; PHI 7 d
Iprovalicarb 5.5% + Propineb 61.25% WP	Early & late blight	2.25-2.5 kg/ha	500 L	V1	CIB&RC 2025; PHI 5 d
Kresoxim-methyl 18% + Mancozeb 54% WP	Early blight, late blight, fruit rot	1388 g/ha	500 L	V1	CIB&RC 2025; PHI 7 d
Pyraclostrobin 10% +	Early & late blight	562.5 g/ha	500 L	V1	CIB&RC 2025; PHI 10 d

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Metiram 30% + Difenoconazole 10% WG					
Trifloxystrobin 10% + Difenoconazole 12.5% + Sulphur 3% SC	Early & late blight	450 ml/ha	500 L	V1	CIB&RC 2025; PHI 14 d
Trifloxystrobin 25% + Hexaconazole 25% WG	Early blight, late blight, powdery mildew, fruit spot	200 g/ha	500 L	V1	CIB&RC 2025; PHI 7 d
Valifenalate 6% + Mancozeb 60% WG	Early & late blight	2000-2500 g/ha	500 L	V1	CIB&RC 2025; PHI 5 d
Flubendiamide 7.5% + Kresoxim-methyl 37.5% SC	Early & late blight	667 ml/ha	500 L	V1	CIB&RC 2025; PHI 7 d
Copper hydroxide 77% WP	Late blight (preventive)	2.0 g/litre, before disease onset	-	V3	ICAR (IH26); no tomato claim in the CIB&RC 2025 list

The disease spreads quickly, so ICAR advises early action with preventive copper sprays before the disease appears (IH26). Cyazofamid and azoxystrobin + difenoconazole, earlier V2 (TNAU), are now V1 (CIB&RC 2025). Recent trade-press reports of Registration Committee meetings mentioned further combinations; the ones that also appear in the CIB&RC 2025 list are included above with printed doses. Check PHI (3-26 days) before the last pre-harvest spray.

6. Sources and citations

[T8] TNAU Agritech Portal - Tomato late blight (Phytophthora infestans). https://agritech.tnau.ac.in/crop_protection/tomato_diseases_8.html

[IH26] Singh, R. (2026). Late blight disease of tomato: a serious problem for Indian tomato farmers. Indian Horticulture, Jan-Feb 2026, pp. 34-35, ICAR-DKMA. https://epubs.icar.org.in/index.php/IndHort/article/download/164935/64599/508183

[RED] Tomato late blight yield loss assessment and risk aversion with resistant hybrid (ICAR-Indian Institute of Horticultural Research authors). https://www.redalyc.org/journal/5770/577073682013/html

[AA] ICAR-IIHR: Multiple disease resistant tomato hybrid Arka Abhed (H-397). https://iihr.res.in/node/23884

[NUN] Nunhems India (company extension page): Tomato disease management - look-alike diseases, begomovirus notes (supplementary, non-university source). https://www.nunhems.com/in/en/disease-management/tomato-disease-management

[UGA] University of Georgia Extension Bulletin B1285: Common tomato diseases in Georgia (buckeye rot vs late blight lesions). https://openscholar.uga.edu/record/21794/files/B1285.pdf

[NCI] NCIPM-based list of tomato pests/diseases and reported states (via Vikaspedia): Tomato pests. https://vikaspedia.in/agriculture/crop-production/integrated-pest-managment/ipm-for-vegetables/ipm-strategies-for-tomato/tomato-pests

[RC464] Trade-press report of 464th CIB&RC Registration Committee meeting (secondary source; confirm against official RC minutes). https://gubbagroup.com/india-approves-27-new-crop-protection-products-under-section-93-at-464th-rc-meeting-2/

[RC465] Trade-press report of 465th CIB&RC Registration Committee meeting (secondary source; confirm against official RC minutes). https://www.global-agriculture.com/crop-protection/indias-465th-registration-committee-meeting-cibrc-clears-19-new-pesticide-registrations/

[CIB] Central Insecticide Board & Registration Committee (CIB&RC), DPPQS, Govt. of India: 'Major Uses of Pesticides (registered under the Insecticides Act, 1968) - Fungicides', up to 30/09/2025 (single products pp. 2-41; combinations pp. 42-79). Copy reviewed (hosted on a university website, not the live register): https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Official portal: http://cibrc.nic.in/ . The compilation states it is for guidance and not for legal purposes.

[CIB12] CIB&RC 'Major Uses of Pesticides - Fungicides' as on 30.09.2012 (older compilation, used for the first version of this document). https://agritech.tnau.ac.in/crop_protection/pdf/6_Major_use_fungicides.pdf

4. Septoria Leaf Spot

Tomato | Septoria lycopersici | Fungal (debris-borne)

At a glance. Many small round spots with grey centres and dark margins on leaves, leading to blighting and defoliation. Spread by rain splash in humid weather.

1. Identification

Symptoms

- Small, round to irregular spots with a grey centre and dark margin on leaves; spots coalesce, leaves blight and may defoliate completely; stems and flowers can be affected (T4).
- Usually begins on the lowest/older leaves, where humidity is highest (UME).
- Tiny black dots (pycnidia) are visible in the grey centres (UME; T4 describes globose pycnidia).

Distinguishing features

- Septoria spots have grey centres with dark margins and lack the concentric rings of early blight (compare T2 and T4).
- Fruit is rarely infected (UME).

2. Causes and progression

- Pathogen: Septoria lycopersici Speg. - hyaline septate mycelium; globose pycnidia; thread-like (filiform) septate pycnidiospores (T4).
- Survival: infected plant debris (T4). Related Solanaceous weeds such as nightshade can also carry the fungus (UME, UCN).
- Spread: rain splash and wind-borne conidia; also equipment and workers when foliage is wet (T4, UCN).
- Progression: spores enter through leaf stomata; black pycnidia form in the spots and release new spores, so infection repeats through the season (UCN).

3. Favourable conditions

- High humidity (T4).
- Extension sources: long periods of leaf wetness and humidity with temperatures of roughly 15-27 C (60-80 F) (UM); optimum about 20-25 C (68-77 F) (UME).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Remove and destroy affected plant parts and crop debris (T4).
- Rotate away from tomato for 2-3 years and control Solanaceous weeds (UCN, UME).
- Reduce leaf wetness as much as possible to slow spread (UCN).
- Seed treatment with Trichoderma asperellum 4 g/kg seed (T4).

Maharashtra note

No Maharashtra-specific recommendation was found for this disease in the sources reviewed.

5. Remedies and treatment

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Azoxystrobin 25% + Boscalid 35% WG	Early blight and Septoria leaf spot	500 g/ha	500 L	V1	CIB&RC 2025; PHI 10 d
Azoxystrobin 4.8% + Chlorothalonil 40% w/w SC	Early blight, late blight, powdery mildew, Septoria leaf blight	3 (ml or g; unit as printed)	500 L	V1	CIB&RC 2025; PHI 3 d
Fluxapyroxad 250 g/l + Pyraclostrobin 250 g/l SC	Early blight and Septoria leaf spot	200-250 ml/ha	500 L	V1	CIB&RC 2025; PHI 10 d
Fluxapyroxad 75 g/l + Difenoconazole 50 g/l SC	Early blight and Septoria leaf spot	800-1000 ml/ha	500 L	V1	CIB&RC 2025; PHI 10 d
Mefentrifluconazole 200 g/l + Pyraclostrobin 200	Early blight and Septoria leaf spot	400-500 ml/ha	500 L	V1	CIB&RC 2025; PHI 7 d

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
g/l SC					
Mancozeb 75% WP	Late blight, buckeye rot, leaf spot (not Septoria-specific)	1.5-2.0 kg/ha	750 L	V1	CIB&RC 2025; PHI not stated
Copper oxychloride 50% WP	Tomato 'leaf spot' (not Septoria-specific)	2.5 kg/ha	750-1000 L	V1	CIB&RC 2025; PHI not stated

Septoria leaf spot now has specific tomato claims in the CIB&RC 2025 list (rows tagged V1 with 'Septoria' in the claim); the earlier TNAU-only fluxapyroxad + pyraclostrobin entry (200-250 ml/ha) is matched. Mancozeb and copper oxychloride are listed for 'leaf spot' in general, not for Septoria by name. Follow the label for interval and PHI.

6. Sources and citations

[T4] TNAU Agritech Portal - Tomato Septoria leaf spot. https://agritech.tnau.ac.in/crop_protection/tomato_diseases_4.html

[UM] UMass Extension: Tomato disease scouting guide for IPM field walk (Septoria, early blight look-alikes). https://ag.umass.edu/sites/ag.umass.edu/files/pdf-doc-ppt/tomato_disease_accessible.pdf

[UME] University of Maine Cooperative Extension: Septoria leaf spot of tomato, Fact sheet #5088. https://extension.umaine.edu/ipm/?p=4626

[UCN] UConn Extension: Problems with Septoria leaf spot on tomatoes. https://ipm.cahnr.uconn.edu/problems-with-septoria-leaf-spot-on-tomatoes-in-2015

[CIB] Central Insecticide Board & Registration Committee (CIB&RC), DPPQS, Govt. of India: 'Major Uses of Pesticides (registered under the Insecticides Act, 1968) - Fungicides', up to 30/09/2025 (single products pp. 2-41; combinations pp. 42-79). Copy reviewed (hosted on a university website, not the live register): https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Official portal: http://cibrc.nic.in/ . The compilation states it is for guidance and not for legal purposes.

[CIB12] CIB&RC 'Major Uses of Pesticides - Fungicides' as on 30.09.2012 (older compilation, used for the first version of this document). https://agritech.tnau.ac.in/crop_protection/pdf/6_Major_use_fungicides.pdf

5. Fusarium Wilt

Tomato | Fusarium oxysporum f. sp. lycopersici | Fungal (soil-borne, vascular)

At a glance. Yellowing and wilting from the lower leaves with brown vascular tissue. Survives in soil for years; resistant varieties, rotation and soil hygiene are the main tools.

1. Identification

Symptoms

- Clearing of the veinlets and yellowing of lower leaves; younger leaves die in succession; petioles and leaves droop and wilt (T3).
- Browning of the vascular system when the stem is cut; the entire plant wilts and dies (T3).
- Often one side or one branch yellows first ('yellow flag') (UCF).

Distinguishing features

- Verticillium wilt looks very similar; Fusarium discoloration tends to be darker and Fusarium prefers warmer soils (UCG).
- Bacterial wilt: sudden wilt with green leaves and white bacterial ooze when a cut stem is placed in water; Fusarium shows progressive yellowing and no ooze (see T5).
- Laboratory culture may be needed to confirm (UCF).

2. Causes and progression

- Pathogen: Fusarium oxysporum f. sp. lycopersici (Sacc.) W.C. Snyder & H.N. Hansen; produces microconidia, macroconidia and chlamydospores (T3). Infects tomato only and exists as three races (UCF).
- Survival: in soil as mycelium and chlamydospores (T3); can persist for more than 10 years (T10).
- Primary spread: infected debris. Secondary: water and planting equipment (T3). Long-distance: seed, transplants and soil on machinery (UCF).
- Progression: the fungus enters through root tips or wounds, invades the xylem and blocks water flow, causing yellowing, wilting and death (UCF, UCG).

3. Favourable conditions

- TNAU (disease page): dry weather (high soil temperature) and low soil moisture (T3).
- TNAU (post-harvest page): optimum temperature about 28 C; alternating low and high soil temperature, high humidity, light sandy soil, low soil moisture and low pH (T10). UC: optimum about 27 C (UCD).
- Root-knot nematode feeding can break resistance and increase wilt (UCF).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Use resistant varieties; some are resistant to all three races (UCF). MPKV Rahuri lists Fusarium-wilt resistance as a breeding objective (MP).
- Crop rotation (T3); rotation reduces inoculum but the fungus is long-lived (UCF).
- Soil solarization; remove and destroy infected plants (T3).
- Clean farm equipment to limit spread of infested soil; manage root-knot nematodes (UCF).
- Raise seedlings in clean (disease-free) soil, since seedlings from infected soil carry the pathogen into the field (T10).

Maharashtra note

NCIPM-based list records Fusarium wilt on tomato in Maharashtra (NCI).

5. Remedies and treatment

Biological / soil-applied options

- Seed treatment with Trichoderma asperellum (viride) 4 g/kg seed (T3).
- Seedling dip in Bacillus subtilis 5 g/litre, plus soil application of B. subtilis 1.0 kg with 50 kg FYM per acre at 30 days after transplanting (T3).
- CIB&RC lists Trichoderma viride 1% WP on tomato for seedling wilt (Fusarium oxysporum); apply per label (CIB).

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Carbendazim	Seed treatment (post-harvest Fusarium rot page)	2 g/kg seed	-	V2	TNAU (T10)

Once plants show wilt, no foliar fungicide cures them. Management is preventive: resistant varieties, clean soil/seedlings, rotation, solarization and biological seed/seedling treatment. The carbendazim seed-treatment dose comes from a TNAU post-harvest page and could not be matched to a tomato claim in the CIB&RC 2025 list, so check the current label.

6. Sources and citations

[T3] TNAU Agritech Portal - Tomato Fusarium wilt. https://agritech.tnau.ac.in/crop_protection/tomato_diseases_3.html

[T10] TNAU Agritech Portal - Post-harvest diseases, Tomato: Fusarium rot (content validated by Dr M. Deivamani, HRS Yercaud). https://agritech.tnau.ac.in/crop_protection/tomato_phdiseases_8.html

[UCF] UC IPM Pest Management Guidelines: Tomato - Fusarium wilt (Fusarium oxysporum f. sp. lycopersici). https://ipm.ucanr.edu/PMG/r783101011.html

[UCG] UC IPM: Fusarium wilt on tomatoes (home garden). https://ipm.ucanr.edu/PMG/GARDEN/VEGES/DISEASES/fusarwilt.html

[UCD] UC Davis / UC ANR: Demystifying Fusarium diseases of tomato (slides; optimum temperature of Fusarium wilt ~27 C). https://ucanr.edu/media/307570

[CIB] Central Insecticide Board & Registration Committee (CIB&RC), DPPQS, Govt. of India: 'Major Uses of Pesticides (registered under the Insecticides Act, 1968) - Fungicides', up to 30/09/2025 (single products pp. 2-41; combinations pp. 42-79). Copy reviewed (hosted on a university website, not the live register): https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Official portal: http://cibrc.nic.in/ . The compilation states it is for guidance and not for legal purposes.

[MP] Mahatma Phule Krishi Vidyapeeth, Rahuri - Tomato Improvement Scheme (objectives, released varieties incl. Phule Raja). https://mpkv.ac.in/Uploads/Research/20260819_123737_3fbf99aa75c9858af9fe7ad9fbe7faaf.pdf

[NCI] NCIPM-based list of tomato pests/diseases and reported states (via Vikaspedia): Tomato pests. https://vikaspedia.in/agriculture/crop-production/integrated-pest-managment/ipm-for-vegetables/ipm-strategies-for-tomato/tomato-pests

6. Bacterial Wilt

Tomato | Ralstonia solanacearum | Bacterial (soil- and water-borne)

At a glance. Rapid wilting of healthy-looking plants with white bacterial ooze from the cut stem. Warm, wet soils. No effective chemical; rely on resistant hybrids, rotation and soil/water management.

1. Identification

Symptoms

- Rapid and complete wilting of plants; lower leaves may drop before wilting (T5).
- Yellow-brown discoloration of tissues; white bacterial ooze from cut ends (T5).
- Leaves usually stay green and attached while the plant wilts (EW; supplementary source).

Distinguishing features

- A freshly cut stem base placed in water streams white, milky bacterial ooze (T5, EW); Fusarium and Verticillium wilts do not.

2. Causes and progression

- Pathogen: Ralstonia solanacearum (Smith) Yabuuchi (T5).
- Survival: infected plants and soil. Primary spread: infected plants/seedlings. Secondary spread: wounds, soil and implements (T5).
- Progression: bacteria enter through roots/wounds, multiply in the xylem and block water movement, producing wilt and death.

3. Favourable conditions

- High soil moisture (T5).
- Study of tomato: incidence rose as soil moisture increased from 40 % to 90 % and soil temperature from 25 to 35 C, with the highest incidence at 30-35 C and 90 % soil moisture (BW1).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Crop rotation, disease-free seedlings, and restricted irrigation (T5).
- Resistant hybrids from ICAR-IIHR: Arka Rakshak, Arka Samrat and Arka Abhed are reported resistant to bacterial wilt (AR, IH88, AA).
- Avoid moving infested soil on tools and seedlings (T5); the disease is common in heavy, low-lying soils that hold moisture (EW).
- Grafting on resistant rootstock: an ICAR project reported complete protection in brinjal grafted on Solanum torvum (BW2); for tomato this should be treated as an option to discuss with the local KVK, not a standard recommendation.

Maharashtra note

NCIPM-based list records bacterial wilt on tomato in Maharashtra (NCI).

5. Remedies and treatment

Biological / soil-applied options

- Soil application of neem cake 100 kg/acre (T5).

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Bleaching powder (soil drench)	Bacterial wilt	4 kg/ac	-	V2	TNAU (T5)

No bactericide is recommended: an open-textbook chapter on tomato diseases states that no chemical has proved effective against the bacterium (COL). Streptocycline is registered by CIB&RC on tomato for BACTERIAL LEAF SPOT only (40-100 ppm sprays; CIB&RC 2025), not for bacterial wilt, so it is not recommended here.

6. Sources and citations

[T5] TNAU Agritech Portal - Tomato bacterial wilt. https://agritech.tnau.ac.in/crop_protection/tomato_diseases_5.html

[BW1] Prevalence and epidemiology of bacterial wilt of tomato (soil temperature 25-35 C and soil moisture 40-90 %). Journal hosted at https://ojs.pphouse.org/index.php/IJEP/article/download/4769/2229

[BW2] D'Souza M. & N.P. (ICAR PhytoFuRa outreach project): Management of bacterial wilt of brinjal using wild brinjal (Solanum torvum) as rootstock. Indian Phytopathology. https://epubs.icar.org.in/index.php/IPPJ/article/view/60167

[AR] Lok Sabha / Parliament reply on ICAR-IIHR hybrid 'Arka Rakshak' (resistance to leaf curl, bacterial wilt, early blight). https://eparlib.nic.in/bitstream/123456789/663314/1/17708.pdf

[IH88] Hanson P. et al. Having a basket full of disease-free tomatoes (Arka Samrat, Arka Rakshak). Indian Horticulture, ICAR. https://epubs.icar.org.in/index.php/IndHort/article/view/88525

[AA] ICAR-IIHR: Multiple disease resistant tomato hybrid Arka Abhed (H-397). https://iihr.res.in/node/23884

[COL] Open textbook 'Integrated Disease Management', Commonwealth of Learning - Ch. 3: Tomato diseases and their management. https://openbooks.col.org/integrateddiseasemanagement/chapter_3-tomato-diseases-and-their-management.html

[EW] East-West Seed Plant Doctor: Bacterial wilt of tomato (company extension page; supplementary, non-university source). https://plantdoctor.eastwestseed.com/node/335

[CIB] Central Insecticide Board & Registration Committee (CIB&RC), DPPQS, Govt. of India: 'Major Uses of Pesticides (registered under the Insecticides Act, 1968) - Fungicides', up to 30/09/2025 (single products pp. 2-41; combinations pp. 42-79). Copy reviewed (hosted on a university website, not the live register): https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Official portal: http://cibrc.nic.in/ . The compilation states it is for guidance and not for legal purposes.

[NCI] NCIPM-based list of tomato pests/diseases and reported states (via Vikaspedia): Tomato pests. https://vikaspedia.in/agriculture/crop-production/integrated-pest-managment/ipm-for-vegetables/ipm-strategies-for-tomato/tomato-pests

7. Bacterial Stem and Fruit Canker

Tomato | Clavibacter michiganensis subsp. michiganensis | Bacterial (seed- and debris-borne)

At a glance. Seed-borne bacterial disease causing leaf scorch, wilting, stem streaks/cankers and 'bird's-eye' fruit spots. Prevention (clean seed, sanitation, rotation) is the main control.

1. Identification

Symptoms

- Seedlings, leaves, stems and fruits can be affected (T9).
- White blister-like spots on leaf margins that turn brown; wilting of leaves and shoots; streaks on stems and petioles (T9).
- Fruit: small water-soaked spots with a white halo; centres become slightly raised, tan and rough (T9).
- Cankers develop and crack with slimy bacterial ooze; vascular discoloration (T9).

Distinguishing features

- Canker is described with vascular discoloration and fruit spots that have a white halo (T9); bacterial spot lesions lack a prominent halo (UM).

2. Causes and progression

- Pathogen: Clavibacter michiganensis subsp. michiganensis (Smith) Davis - Gram-positive, non-motile, rod-shaped, aerobic bacterium (T9).
- Survival: infected debris. Primary spread: infected plant debris and seed. Secondary spread: rain splash (T9).

3. Favourable conditions

- Soil temperature around 28 C (T9).
- High humidity or persistent dew; moist weather with intermittent showers (T9).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Use disease-free seed and healthy transplants (T9).
- Field sanitation and removal of weeds; crop rotation with non-host crops (T9).
- Disinfect tools and implements (T9).

Maharashtra note

NCIPM-based list records bacterial stem and fruit canker on tomato in Maharashtra (NCI).

5. Remedies and treatment

Chemical options with verification status

No chemical recommendation is given (tag --).

No chemical recommendation is given. TNAU lists only cultural measures for this disease (T9). The CIB&RC 2025 list has tomato bactericide claims for BACTERIAL LEAF SPOT (streptomycin + tetracycline 40-100 ppm; kasugamycin combinations), not for canker. If a chemical is needed, consult the local KVK/agricultural university and verify the label claim on the CIB&RC register.

6. Sources and citations

[T9] TNAU Agritech Portal - Tomato bacterial stem and fruit canker. https://agritech.tnau.ac.in/crop_protection/tomato_diseases_9.html

[UM] UMass Extension: Tomato disease scouting guide for IPM field walk (Septoria, early blight look-alikes). https://ag.umass.edu/sites/ag.umass.edu/files/pdf-doc-ppt/tomato_disease_accessible.pdf

[NCI] NCIPM-based list of tomato pests/diseases and reported states (via Vikaspedia): Tomato pests. https://vikaspedia.in/agriculture/crop-production/integrated-pest-managment/ipm-for-vegetables/ipm-strategies-for-tomato/tomato-pests

[CIB] Central Insecticide Board & Registration Committee (CIB&RC), DPPQS, Govt. of India: 'Major Uses of Pesticides (registered under the Insecticides Act, 1968) - Fungicides', up to 30/09/2025 (single products pp. 2-41; combinations pp. 42-79). Copy reviewed (hosted on a university website, not the live register): https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Official portal: http://cibrc.nic.in/ . The compilation states it is for guidance and not for legal purposes.

8. Tomato Leaf Curl Disease

Tomato | Tomato leaf curl virus (Begomovirus); vector: whitefly Bemisia tabaci | Viral (whitefly-transmitted)

At a glance. Stunting with downward rolling and crinkling of leaves, spread by whiteflies. One of the most damaging tomato diseases in India. Vector control, resistant hybrids and protected nurseries are key.

1. Identification

Symptoms

- Severe stunting of plants; downward rolling and crinkling of leaves (T6).
- Older leaves become leathery and brittle; shortened internodes; more lateral branches giving a bushy appearance (T6).
- Other described symptoms: vein clearing, reduced leaf size and leaflet puckering (TL2).

Distinguishing features

- Symptoms vary with cultivar, temperature, mixed infections and virus species (NUN, TL2).
- Tomato spotted wilt/bud necrosis shows ring spots and bronzing, not leaf curling (see T7).

2. Causes and progression

- Causal agent: Tomato leaf curl virus; in India several begomovirus species cause the disease alone or together (T6, NUN, TL2).
- Vector: whitefly Bemisia tabaci. Primary spread: whiteflies. Secondary spread: movement of infected plants (T6).
- The disease is not transmitted through seed, soil or by handling; spread follows whitefly movement (NUN).
- Reservoir: cultivated crops and weeds near fields; TNAU names Abutilon indicum as an alternate weed host (T6, NUN).

3. Favourable conditions

- Incidence occurs throughout the year but varies with season, temperature, humidity and rainfall, and is strongly driven by whitefly numbers (TL2).
- Whitefly population correlates positively with temperature and sunshine hours; cooler weather, high humidity and rainfall reduce whitefly build-up (TL2).
- An Indian study recorded 83 % leaf curl in an October (winter) planting against 14 % in a February (summer) planting, linked to whitefly numbers at early crop stages (TL3). Another study reports whitefly populations were higher in December-February when temperatures were low (TL1); findings differ by location and season, so follow local whitefly monitoring.

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Raise the nursery under protected conditions (insect-proof net); use disease-free seedlings (T6).
- Barrier crop of sorghum, maize or bajra around the field (T6).
- Yellow sticky traps at 5 per acre; remove alternate weed hosts such as Abutilon indicum (T6).
- Mulches: polythene mulch lowered incidence in an Indian trial (TL3); UV-reflective and paddy-husk mulch gave the lowest incidence in a trial at Hyderabad (TL4).
- Resistant hybrids: Arka Rakshak, Arka Samrat (AR, IH88) and Arka Abhed (AA) (IIHR). For Maharashtra, MPKV Rahuri's variety Phule Raja is described as tolerant to thrips, whitefly, leaf curl and spotted wilt virus (MP).
- Planting date strongly affected incidence in an Indian trial (83 % vs 14 %), so avoid sowing into periods of high whitefly pressure in your locality (TL3).

Maharashtra note

NCIPM-based list records tomato leaf curl virus in Maharashtra (NCI). MPKV Rahuri targets leaf curl resistance in its tomato breeding (MP).

5. Remedies and treatment

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Thiamethoxam 25% WG	Whitefly vector	200 g/ha (50 g a.i.)	500 L	V1	CIB&RC insecticide list (CIBI); PHI 5 d
Imidacloprid 70% WG	Thrips and whitefly (Bemisia tabaci)	50 g/ha (35 g a.i.)	500 L	V1	CIB&RC insecticide list (CIBI, 2021 version); PHI 5 d
Imidacloprid (SL / SC formulation)	Whitefly vector	about 150-175 ml/ha formulation (30-35 g a.i.); the formulation heading in the source text is unclear, so check the exact product label	500 L	V1	CIB&RC insecticide list (CIBI); PHI 3 d
Cyantraniliprole 10.26% OD	Whitefly, thrips (tomato)	900 ml/ha (90 g a.i.)	500 L	V1	CIB&RC insecticide list (CIBI); PHI 3 d
Dimethoate 30 EC	Whitefly vector	TNAU: 200 ml/ac (not matched to a tomato claim in the CIB&RC lists reviewed)	-	V2	TNAU (T6) only
Methyl demeton 25 EC	Whitefly vector	TNAU: 200 ml/ac (not matched to a tomato claim in the CIB&RC lists reviewed)	-	V2	TNAU (T6) only

These insecticides target the VECTOR, not the virus; infected plants do not recover. The V1 rows come from older CIB&RC insecticide lists (2019 / 2021); the 2025 insecticide list was not retrieved, so confirm the current tomato label claim, PHI and regulatory status. Cross-check: the earlier TNAU entries for thiamethoxam 25 WG (200 g/ac) and imidacloprid 17.8 SL (200 ml/ac) are numerically the same as the CIB&RC per-hectare doses (200 g/ha; about 150-175 ml/ha), so applying the TNAU per-acre figure would give roughly 2.5 times the label dose. Use the CIB&RC/label dose. Dimethoate and methyl demeton remain TNAU-only. Rotate insecticide groups to delay resistance, and combine with the non-chemical measures above.

6. Sources and citations

[T6] TNAU Agritech Portal - Tomato leaf curl (Tomato leaf curl virus). https://agritech.tnau.ac.in/crop_protection/tomato_diseases_6.html (updated Jun 2023)

[TL1] Singh R.K. et al. (2015). Effect of climate change on tomato leaf curl virus (ToLCV) disease in tomato. Indian Journal of Agricultural Sciences 85(2). https://epubs.icar.org.in/index.php/IJAgS/article/download/46547/20134/377767

[TL2] Agricultural Reviews (ARCC Journals): review on tomato leaf curl disease, R-2346 (whitefly vs temperature/humidity; begomovirus species in India). https://arccjournals.com/journal/agricultural-reviews/R-2346

[TL3] Eco-friendly management of leaf curl disease of tomato. Indian Phytopathology (Indian Phytopathological Society). https://epubs.icar.org.in/index.php/IPPJ/article/view/18773

[TL4] Rajasri M. et al. Effect of different mulches on the incidence of tomato leaf curl virus and its vector whitefly Bemisia tabaci in tomato (ANGRAU, Hyderabad). Acta Horticulturae 914. https://ishs.org/ishs-article/914_39/

[NUN] Nunhems India (company extension page): Tomato disease management - look-alike diseases, begomovirus notes (supplementary, non-university source). https://www.nunhems.com/in/en/disease-management/tomato-disease-management

[AR] Lok Sabha / Parliament reply on ICAR-IIHR hybrid 'Arka Rakshak' (resistance to leaf curl, bacterial wilt, early blight). https://eparlib.nic.in/bitstream/123456789/663314/1/17708.pdf

[IH88] Hanson P. et al. Having a basket full of disease-free tomatoes (Arka Samrat, Arka Rakshak). Indian Horticulture, ICAR. https://epubs.icar.org.in/index.php/IndHort/article/view/88525

[AA] ICAR-IIHR: Multiple disease resistant tomato hybrid Arka Abhed (H-397). https://iihr.res.in/node/23884

[MP] Mahatma Phule Krishi Vidyapeeth, Rahuri - Tomato Improvement Scheme (objectives, released varieties incl. Phule Raja). https://mpkv.ac.in/Uploads/Research/20260819_123737_3fbf99aa75c9858af9fe7ad9fbe7faaf.pdf

[NCI] NCIPM-based list of tomato pests/diseases and reported states (via Vikaspedia): Tomato pests. https://vikaspedia.in/agriculture/crop-production/integrated-pest-managment/ipm-for-vegetables/ipm-strategies-for-tomato/tomato-pests

[RC464] Trade-press report of 464th CIB&RC Registration Committee meeting (secondary source; confirm against official RC minutes). https://gubbagroup.com/india-approves-27-new-crop-protection-products-under-section-93-at-464th-rc-meeting-2/

[CIBI] CIB&RC 'Major Uses of Pesticides - Insecticides' (older versions, as on 31.10.2019 and 01.01.2021, hosted by the Odisha agriculture department; the 2025 insecticide list was not retrieved). https://www.epestodisha.nic.in/public/documents/Major%20use%20of%20Pesticides%20by%20CIBRC.pdf and https://www.epestodisha.nic.in/public/documents/Major%20Use%20as%20on%201.01.2021.pdf

9. Tomato Spotted Wilt / Bud Necrosis

Tomato | Tomato spotted wilt virus / Groundnut bud necrosis virus (Orthotospovirus); vector: thrips | Viral (thrips-transmitted)

At a glance. Stunting, ring spots and bronzing of young leaves, tip die-back and ring-marked fruit, spread by thrips. Healthy seedlings, early roguing, barrier crops and vector control are used.

1. Identification

Symptoms

- Severe stunting; chlorotic or necrotic rings on leaves (T7).
- Thickening of veins and bronzing of young leaves; growing tips die back and wilt (T7).
- Fruit shows yellow areas with concentric circular markings; seed discoloration (T7).

Distinguishing features

- Ring spots and bronzing/tip necrosis distinguish it from leaf curl, where leaves roll and crinkle.

2. Causes and progression

- Causal agent in TNAU: Groundnut bud necrosis virus / Tomato spotted wilt virus (T7); NCIPM lists a Peanut bud necrosis virus (TSWV group) form on tomato (NCI).
- Vectors: thrips - TNAU names Frankliniella schultzii and Scirtothrips dorsalis (T7).
- Primary spread: infected plants. Secondary spread: thrips (T7). Weeds and other hosts act as reservoirs, so collateral-host management matters (COL).

3. Favourable conditions

- The TNAU page does not give temperature or humidity thresholds. Disease follows thrips activity and the presence of infected plants and alternate hosts (T7, COL).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Use healthy seedlings for transplanting (T7).
- Rogue (remove) infected plants up to 45 days after planting (T7).
- Grow sorghum, maize or bajra as a barrier crop (T7).
- Manage weeds and other collateral hosts (COL).
- MPKV Rahuri targets spotted wilt resistance in tomato breeding; Phule Raja is described as tolerant to thrips, whitefly, leaf curl and spotted wilt (MP).

Maharashtra note

In the NCIPM-based list, tomato spotted wilt/bud necrosis is recorded for Himachal Pradesh, Andhra Pradesh, Karnataka and Tamil Nadu, not Maharashtra (NCI); MPKV still lists spotted wilt resistance as a breeding target (MP).

5. Remedies and treatment

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Cyantraniliprole 10.26% OD	Thrips / whitefly on tomato	900 ml/ha (90 g a.i.); TNAU: 900 ml/ha	500 L	V1	CIB&RC insecticide list (CIBI); PHI 3 d
Imidacloprid 70% WG	Thrips and whitefly on tomato	50 g/ha (35 g a.i.)	500 L	V1	CIB&RC insecticide list (CIBI, 2021 version); PHI 5 d
Thiamethoxam 70% WS	Aphids & thrips on tomato	TNAU lists 300 ml/ha; CIB&RC lists thiamethoxam 70 WS on tomato only as a seed dresser at sowing	-	V1	CIB&RC insecticide list (CIBI): seed dresser only
Imidacloprid 17.8% SL	Thrips vector	TNAU: 250 ml/ha (not matched to a tomato thrips claim in the CIB&RC lists reviewed)	-	V2	TNAU (T7) only

TNAU says to apply any one of these. CIB&RC lists cyantraniliprole 10.26% OD for tomato thrips/whitefly and imidacloprid 70% WG for thrips and whitefly; thiamethoxam 70% WS is registered on tomato as a seed dresser, not a foliar spray, so the TNAU foliar entry does not match the label. As with leaf curl, these act on

the vector. Insecticide entries come from older CIB&RC versions; confirm label claim, PHI and current status before use.

6. Sources and citations

[T7] TNAU Agritech Portal - Tomato spotted wilt disease / Groundnut bud necrosis virus. https://agritech.tnau.ac.in/crop_protection/tomato_diseases_7.html

[COL] Open textbook 'Integrated Disease Management', Commonwealth of Learning - Ch. 3: Tomato diseases and their management. https://openbooks.col.org/integrateddiseasemanagement/chapter_3-tomato-diseases-and-their-management.html

[NCI] NCIPM-based list of tomato pests/diseases and reported states (via Vikaspedia): Tomato pests. https://vikaspedia.in/agriculture/crop-production/integrated-pest-managment/ipm-for-vegetables/ipm-strategies-for-tomato/tomato-pests

[MP] Mahatma Phule Krishi Vidyapeeth, Rahuri - Tomato Improvement Scheme (objectives, released varieties incl. Phule Raja). https://mpkv.ac.in/Uploads/Research/20260819_123737_3fbf99aa75c9858af9fe7ad9fbe7faaf.pdf

[CIBI] CIB&RC 'Major Uses of Pesticides - Insecticides' (older versions, as on 31.10.2019 and 01.01.2021, hosted by the Odisha agriculture department; the 2025 insecticide list was not retrieved). https://www.epestodisha.nic.in/public/documents/Major%20use%20of%20Pesticides%20by%20CIBRC.pdf and https://www.epestodisha.nic.in/public/documents/Major%20Use%20as%20on%201.01.2021.pdf

10. Bacterial Leaf Spot (Bacterial Spot)

Tomato | Xanthomonas axonopodis pv. vesicatoria | Bacterial (seed- and debris-borne; rain-splashed)

At a glance. Small water-soaked leaf spots with a yellow halo, scabby fruit spots and defoliation of older leaves, spread by rain splash. TNAU's recommendation is disease-free seed plus streptocycline sprays; CIB&RC lists a tomato bacterial leaf spot claim.

1. Identification

Symptoms

- Small, brown, water-soaked, circular spots with a yellow halo on leaves; older leaves may defoliate (T11).
- Small, water-soaked spots on green fruit; the centres become irregular, light brown, sunken and scabby (T11).

Distinguishing features

- Bacterial spot lesions have no concentric zones like early blight (UM).
- Another extension source states bacterial spot lesions have no prominent halo (UM), whereas TNAU describes a yellow halo (T11); confirm in the lab if unsure.

2. Causes and progression

- Pathogen: Xanthomonas axonopodis pv. vesicatoria (T11).
- Survival: infected debris. Primary spread: infected plant debris and seed. Secondary spread: rain splash (T11).

3. Favourable conditions

- Moist weather and splattering rains; high humidity or persistent dew (T11).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Use disease-free seed (T11).
- Because the bacterium survives in infected debris and spreads by rain splash (T11), removing crop debris and limiting leaf wetness are logical sanitation steps; TNAU itself lists only seed and spray measures.

Maharashtra note

No Maharashtra-specific recommendation was found for this disease in the sources reviewed.

5. Remedies and treatment

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Streptomycin sulphate + Tetracycline hydrochloride SP ('Streptocycline')	Bacterial leaf spot	40-100 ppm spray solution (TNAU: 20 g/ac)	-	V1	CIB&RC 2025 (CIB); TNAU (T11)
Thiophanate-methyl 38% + Kasugamycin 2.21% SC	Powdery mildew and bacterial leaf spot	1000-1250 ml/ha (as read from the compilation)	500 L	V1	CIB&RC 2025; PHI 7 d

The antibiotic is used at a solution strength (ppm). Confirm the label for the exact formulation, interval and PHI, and note that repeated antibiotic sprays are limited by resistance concerns. The combination row was read from a printed compilation with clear numbers but check the product label. No chemical is a cure once lesions are established; the aim is protection of healthy tissue.

6. Sources and citations

[T11] TNAU Agritech Portal - Tomato bacterial leaf spot (Xanthomonas axonopodis pv. vesicatoria). Source cited by TNAU: Crop Production Guide - Horticulture 2020 (updated Jun 2023). https://agritech.tnau.ac.in/crop_protection/tomato_diseases_11.html

[UM] UMass Extension: Tomato disease scouting guide for IPM field walk (Septoria, early blight look-alikes). https://ag.umass.edu/sites/ag.umass.edu/files/pdf-doc-ppt/tomato_disease_accessible.pdf

[CIB] Central Insecticide Board & Registration Committee (CIB&RC), DPPQS, Govt. of India: 'Major Uses of Pesticides (registered under the Insecticides Act, 1968) - Fungicides', up to 30/09/2025 (single products pp. 2-41; combinations pp. 42-79). Copy reviewed (hosted on a university website, not the live register): https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Official portal: http://cibrc.nic.in/ . The compilation states it is for guidance and not for legal purposes.

11. Leaf Mould

Tomato | Passalora fulva (Cladosporium fulvum); Pseudocercospora fuligena also reported | Fungal (debris-, seed- and wind-borne; mainly protected cultivation)

At a glance. A foliar disease of tomato, especially in humid polyhouses and high tunnels. Yellow patches on the upper leaf surface with velvety olive mould underneath. Airflow, low humidity, sanitation and resistant varieties are the controls.

1. Identification

Symptoms

- The oldest (lower) leaves are infected first; pale green-yellow spots on the upper leaf side eventually turn yellow, and a velvety mould forms on the underside opposite the spots (LM1, LM3).
- Infected blossoms turn black and fall; fruit can show bronze or sunken leathery lesions but this is uncommon (LM1, LM2).
- Leaves eventually drop prematurely (LM2).

Distinguishing features

- Magnesium deficiency can look similar, but leaf-mould spots are more discrete and the mould appears underneath (LM3).
- Botrytis grey mould forms dark grey mycelia that can resemble the mould of leaf mould (LM2).

2. Causes and progression

- Pathogen: Passalora fulva (syn. Cladosporium fulvum); it infects only tomato (LM1). Leaf mould can also be caused by a second fungus, Pseudocercospora fuligena (LM4).
- Survival: on infected plant debris in soil; spores spread by wind, water and tools, and can enter on infected seed (LM1). It may overwinter as conidia or sclerotia on debris, seed or soil (LM2).
- The pathogen mutates quickly and has many races (LM2).

3. Favourable conditions

- Most prevalent at relative humidity above 85 %, over a broad temperature range of about 10-32 C (50-90 F) (LM1).
- Especially problematic in greenhouses and high tunnels (LM1, LM3).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Remove all crop residue at the end of the season (LM1).
- Use drip irrigation and avoid overhead irrigation; allow airflow through proper plant spacing, staking and pruning (LM1).
- Use varieties with partial or high resistance to leaf mould; resistance can be overcome by new races (LM2, LM4).
- Use clean seed; sanitising seed with dilute bleach is described for seed-borne pathogens (LM2).
- Prune out and destroy symptomatic leaves at the first signs (LM2).

Maharashtra note

No Maharashtra-specific data were found. The disease is most relevant to polyhouse tomato.

5. Remedies and treatment

Chemical options with verification status

Product	Use / target	Dose (formulation)	Water	Tag	Basis / PHI
Zineb 75% WP	Tomato 'grey leaf mould' (claim name as printed; may not be specific to Passalora)	1.5-2.0 kg/ha	750-1000 L	V1	CIB&RC 2025; PHI not stated

This is the only tomato claim found in the CIB&RC 2025 list under a 'leaf mould' name. Check the label to see which pathogen is claimed. Cultural measures that lower humidity matter more than sprays in protected cultivation.

6. Sources and citations

[LM1] Utah State University Extension: Tomato leaf mold (Passalora fulva). https://extension.usu.edu/vegetableguide/tomato-pepper-eggplant/leaf-mold

[LM2] Ohio State University (High tunnel disease facts): Tomato diseases - leaf mold. https://u.osu.edu/hightunneldiseasefacts/?p=146

[LM3] Cornell AgriTech / Cornell Vegetables: Tomato leaf mold. https://vegetables.cornell.edu/crops/tomatoes/tomato-leaf-mold

[LM4] UF/IFAS Extension PP368: Common tomato high-tunnel production diseases in Florida (leaf mold caused by Passalora fulva and Pseudocercospora fuligena). https://edis.ifas.ufl.edu/publication/PP368/pdf

[CIB] Central Insecticide Board & Registration Committee (CIB&RC), DPPQS, Govt. of India: 'Major Uses of Pesticides (registered under the Insecticides Act, 1968) - Fungicides', up to 30/09/2025 (single products pp. 2-41; combinations pp. 42-79). Copy reviewed (hosted on a university website, not the live register): https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Official portal: http://cibrc.nic.in/ . The compilation states it is for guidance and not for legal purposes.

12. Tomato Mosaic Virus (and related mosaic viruses)

Tomato | Tomato mosaic virus (ToMV) and tomato mottle mosaic virus (ToMMV) - tobamoviruses; cucumber mosaic virus (CMV) is also blamed for mosaic in Maharashtra | Viral (seed-, contact- and sap-borne; CMV aphid-borne)

At a glance. Mottled green-yellow mosaic, distorted leaves and small or deformed fruit. Tobamoviruses spread by seed, hands, tools and weeds, so clean seed, sanitation and resistant varieties are the main controls. No chemical cure.

1. Identification

Symptoms

- ToMV: green mottling and yellowing of leaves, which may blister or form fern-like patterns (MV3).
- ToMMV in India: mild mosaic that progresses to severe leaf shoestring, with small, hard, unripe or deformed fruit; more than 50 % of plants showed symptoms in an experimental field (MV1).
- Symptoms depend on variety, plant age at infection and virus strain; on susceptible cultivars symptoms range from severe to none (MV2).

Distinguishing features

- Leaf curl disease rolls and crinkles leaves downward and is whitefly-borne (T6); mosaic shows mottling and shoestring (MV1).
- Seed- and touch-borne spread of tobamoviruses contrasts with CMV, which aphids transmit within minutes (MV3).

2. Causes and progression

- Pathogen: Tomato mosaic virus is a tobamovirus closely related to tobacco mosaic virus (MV2). ToMMV is another tobamovirus, found on tomato in India and present on seed intercepted from India (MV1, MV4).
- Transmission: ToMV is transmitted mechanically (sap, contact), by grafting, by insects to a small extent, and by seed; infected weeds can bring the virus into a field, and the virus in guttation fluid helps workers spread it (MV2). ToMMV is seed-transmissible in a small percentage of seedlings (MV1).
- Press reports from 2023 attributed mosaic crop losses to CMV in Maharashtra and to ToMV in Karnataka and other southern states; CMV is aphid-borne and also seed- and sap-transmissible (MV3, secondary source).

3. Favourable conditions

- The sources reviewed give no temperature or humidity thresholds. Spread follows handling and transplanting: infected seed, saplings, tools and human contact (MV3).

4. Prevention and management

Cultural practices, sanitation, resistant varieties and preventive measures

- Use clean, virus-free seed and healthy saplings; MDPI authors stress reliable diagnosis and management of the virus in tomato seed and nurseries (MV1, MV3).
- Biosafety standards in nurseries and compulsory seed testing are recommended in the press explainer (MV3).
- Remove infected weeds and plants and avoid handling healthy plants after infected ones, because contact and tools spread the virus (MV2).
- Resistance: ToMMV was shown to overcome resistance of tomato cultivars that resist ToMV (MV5); do not rely on ToMV resistance alone.
- Avoid aphid migration into nurseries to limit CMV (MV3) and remove infected weeds (MV2).

Maharashtra note

Press reports link Maharashtra mosaic losses to CMV (MV3). ICAR institutes report ToMMV-type shoestring disease on tomato in India (MV1). No state-wise incidence data were retrieved.

5. Remedies and treatment

Chemical options with verification status

No chemical recommendation is given (tag --).

No chemical recommendation is given (tag --). No tomato claim for any tobamovirus or mosaic virus was found in the CIB&RC 2025 fungicide list, and viruses are not cured by fungicides. For aphid vector control consult the CIB&RC insecticide list and the local KVK; no aphid dose is given here.

6. Sources and citations

[MV1] Biological and molecular characterization of a new isolate of Tomato mottle mosaic virus causing severe shoestring and fruit deformities in tomato plants in India. Plants 2024, 13(19): 2811 (ICAR-NRRI, ICAR-IISR). https://www.mdpi.com/2223-7747/13/19/2811

[MV2] Tomato mosaic virus: transmission studies (Uttar Pradesh, India). Journal of Pharmacognosy and Phytochemistry 6(6) (2017). https://www.phytojournal.com/archives/2017/vol6issue6/PartG/6-5-135-757.pdf

[MV3] Drishti IAS (news explainer, secondary source): CMV and ToMV viruses in tomato crop in Maharashtra and Karnataka. https://www.drishtiias.com/pdf/1688722698.pdf

[MV4] EPPO Global Database: Tomato mottle mosaic virus - distribution in India (present, no details; intercepted on seeds from India). https://gd.eppo.int/taxon/TOMMV0/distribution/IN

[MV5] EPPO: Pest risk analysis quick scan for Tomato mottle mosaic virus (notes that ToMMV has been shown to break ToMV resistance). https://pra.eppo.int/getfile/531404a5-dcde-4d23-a2d6-cd09f49f0b5a

[T6] TNAU Agritech Portal - Tomato leaf curl (Tomato leaf curl virus). https://agritech.tnau.ac.in/crop_protection/tomato_diseases_6.html (updated Jun 2023)

[CIB] Central Insecticide Board & Registration Committee (CIB&RC), DPPQS, Govt. of India: 'Major Uses of Pesticides (registered under the Insecticides Act, 1968) - Fungicides', up to 30/09/2025 (single products pp. 2-41; combinations pp. 42-79). Copy reviewed (hosted on a university website, not the live register): https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Official portal: http://cibrc.nic.in/ . The compilation states it is for guidance and not for legal purposes.

Consolidated source list

[T1] TNAU Agritech Portal - Tomato damping off (Pythium aphanidermatum). Source cited by TNAU: Crop Production Guide - Horticulture 2020. https://agritech.tnau.ac.in/crop_protection/tomato_diseases_1.html

[COL] Open textbook 'Integrated Disease Management', Commonwealth of Learning - Ch. 3: Tomato diseases and their management. https://openbooks.col.org/integrateddiseasemanagement/chapter_3-tomato-diseases-and-their-management.html

[T0] TNAU Agritech Portal - Tomato: crop protection (disease index). https://agritech.tnau.ac.in/crop_protection/crop_prot_crop%20diseases_veg_tomato.html (updated Jun 2023)

[CIB] Central Insecticide Board & Registration Committee (CIB&RC), DPPQS, Govt. of India: 'Major Uses of Pesticides (registered under the Insecticides Act, 1968) - Fungicides', up to 30/09/2025 (single products pp. 2-41; combinations pp. 42-79). Copy reviewed (hosted on a university website, not the live register): https://new.jau.in/storage/document/dynamic_document//residue_fungicide_30092025_20251226_781.pdf . Official portal: http://cibrc.nic.in/ . The compilation states it is for guidance and not for legal purposes.

[CIB12] CIB&RC 'Major Uses of Pesticides - Fungicides' as on 30.09.2012 (older compilation, used for the first version of this document). https://agritech.tnau.ac.in/crop_protection/pdf/6_Major_use_fungicides.pdf

[T2] TNAU Agritech Portal - Tomato early blight (Alternaria solani). https://agritech.tnau.ac.in/crop_protection/tomato_diseases_2.html

[AA] ICAR-IIHR: Multiple disease resistant tomato hybrid Arka Abhed (H-397). https://iihr.res.in/node/23884

[AR] Lok Sabha / Parliament reply on ICAR-IIHR hybrid 'Arka Rakshak' (resistance to leaf curl, bacterial wilt, early blight). https://eparlib.nic.in/bitstream/123456789/663314/1/17708.pdf

[IH88] Hanson P. et al. Having a basket full of disease-free tomatoes (Arka Samrat, Arka Rakshak). Indian Horticulture, ICAR. https://epubs.icar.org.in/index.php/IndHort/article/view/88525

[IH82] Singh, R.K. & Singh, S. (2019). Managing early blight disease: hindrance to tomato fruit production. Indian Horticulture 61(2). https://epubs.icar.org.in/index.php/IndHort/article/view/88382

[NUN] Nunhems India (company extension page): Tomato disease management - look-alike diseases, begomovirus notes (supplementary, non-university source). https://www.nunhems.com/in/en/disease-management/tomato-disease-management

[UM] UMass Extension: Tomato disease scouting guide for IPM field walk (Septoria, early blight look-alikes). https://ag.umass.edu/sites/ag.umass.edu/files/pdf-doc-ppt/tomato_disease_accessible.pdf

[NCI] NCIPM-based list of tomato pests/diseases and reported states (via Vikaspedia): Tomato pests. https://vikaspedia.in/agriculture/crop-production/integrated-pest-managment/ipm-for-vegetables/ipm-strategies-for-tomato/tomato-pests

[MP] Mahatma Phule Krishi Vidyapeeth, Rahuri - Tomato Improvement Scheme (objectives, released varieties incl. Phule Raja). https://mpkv.ac.in/Uploads/Research/20260819_123737_3fbf99aa75c9858af9fe7ad9fbe7faaf.pdf

[T8] TNAU Agritech Portal - Tomato late blight (Phytophthora infestans). https://agritech.tnau.ac.in/crop_protection/tomato_diseases_8.html

[IH26] Singh, R. (2026). Late blight disease of tomato: a serious problem for Indian tomato farmers. Indian Horticulture, Jan-Feb 2026, pp. 34-35, ICAR-DKMA. https://epubs.icar.org.in/index.php/IndHort/article/download/164935/64599/508183

[RED] Tomato late blight yield loss assessment and risk aversion with resistant hybrid (ICAR-Indian Institute of Horticultural Research authors). https://www.redalyc.org/journal/5770/577073682013/html

[UGA] University of Georgia Extension Bulletin B1285: Common tomato diseases in Georgia (buckeye rot vs late blight lesions). https://openscholar.uga.edu/record/21794/files/B1285.pdf

[RC464] Trade-press report of 464th CIB&RC Registration Committee meeting (secondary source; confirm against official RC minutes). https://gubbagroup.com/india-approves-27-new-crop-protection-products-under-section-93-at-464th-rc-meeting-2/

[RC465] Trade-press report of 465th CIB&RC Registration Committee meeting (secondary source; confirm against official RC minutes). https://www.global-agriculture.com/crop-protection/indias-465th-registration-committee-meeting-cibrc-clears-19-new-pesticide-registrations/

[T4] TNAU Agritech Portal - Tomato Septoria leaf spot. https://agritech.tnau.ac.in/crop_protection/tomato_diseases_4.html

[UME] University of Maine Cooperative Extension: Septoria leaf spot of tomato, Fact sheet #5088. https://extension.umaine.edu/ipm/?p=4626

[UCN] UConn Extension: Problems with Septoria leaf spot on tomatoes. https://ipm.cahnr.uconn.edu/problems-with-septoria-leaf-spot-on-tomatoes-in-2015

[T3] TNAU Agritech Portal - Tomato Fusarium wilt. https://agritech.tnau.ac.in/crop_protection/tomato_diseases_3.html

[T10] TNAU Agritech Portal - Post-harvest diseases, Tomato: Fusarium rot (content validated by Dr M. Deivamani, HRS Yercaud). https://agritech.tnau.ac.in/crop_protection/tomato_phdiseases_8.html

[UCF] UC IPM Pest Management Guidelines: Tomato - Fusarium wilt (Fusarium oxysporum f. sp. lycopersici). https://ipm.ucanr.edu/PMG/r783101011.html

[UCG] UC IPM: Fusarium wilt on tomatoes (home garden). https://ipm.ucanr.edu/PMG/GARDEN/VEGES/DISEASES/fusarwilt.html

[UCD] UC Davis / UC ANR: Demystifying Fusarium diseases of tomato (slides; optimum temperature of Fusarium wilt ~27 C). https://ucanr.edu/media/307570

[T5] TNAU Agritech Portal - Tomato bacterial wilt. https://agritech.tnau.ac.in/crop_protection/tomato_diseases_5.html

[BW1] Prevalence and epidemiology of bacterial wilt of tomato (soil temperature 25-35 C and soil moisture 40-90 %). Journal hosted at https://ojs.pphouse.org/index.php/IJEP/article/download/4769/2229

[BW2] D'Souza M. & N.P. (ICAR PhytoFuRa outreach project): Management of bacterial wilt of brinjal using wild brinjal (Solanum torvum) as rootstock. Indian Phytopathology. https://epubs.icar.org.in/index.php/IPPJ/article/view/60167

[EW] East-West Seed Plant Doctor: Bacterial wilt of tomato (company extension page; supplementary, non-university source). https://plantdoctor.eastwestseed.com/node/335

[T9] TNAU Agritech Portal - Tomato bacterial stem and fruit canker. https://agritech.tnau.ac.in/crop_protection/tomato_diseases_9.html

[T6] TNAU Agritech Portal - Tomato leaf curl (Tomato leaf curl virus). https://agritech.tnau.ac.in/crop_protection/tomato_diseases_6.html (updated Jun 2023)

[TL1] Singh R.K. et al. (2015). Effect of climate change on tomato leaf curl virus (ToLCV) disease in tomato. Indian Journal of Agricultural Sciences 85(2). https://epubs.icar.org.in/index.php/IJAgS/article/download/46547/20134/377767

[TL2] Agricultural Reviews (ARCC Journals): review on tomato leaf curl disease, R-2346 (whitefly vs temperature/humidity; begomovirus species in India). https://arccjournals.com/journal/agricultural-reviews/R-2346

[TL3] Eco-friendly management of leaf curl disease of tomato. Indian Phytopathology (Indian Phytopathological Society). https://epubs.icar.org.in/index.php/IPPJ/article/view/18773

[TL4] Rajasri M. et al. Effect of different mulches on the incidence of tomato leaf curl virus and its vector whitefly Bemisia tabaci in tomato (ANGRAU, Hyderabad). Acta Horticulturae 914. https://ishs.org/ishs-article/914_39/

[CIBI] CIB&RC 'Major Uses of Pesticides - Insecticides' (older versions, as on 31.10.2019 and 01.01.2021, hosted by the Odisha agriculture department; the 2025 insecticide list was not retrieved). https://www.epestodisha.nic.in/public/documents/Major%20use%20of%20Pesticides%20by%20CIBRC.pdf and https://www.epestodisha.nic.in/public/documents/Major%20Use%20as%20on%201.01.2021.pdf

[T7] TNAU Agritech Portal - Tomato spotted wilt disease / Groundnut bud necrosis virus. https://agritech.tnau.ac.in/crop_protection/tomato_diseases_7.html

[T11] TNAU Agritech Portal - Tomato bacterial leaf spot (Xanthomonas axonopodis pv. vesicatoria). Source cited by TNAU: Crop Production Guide - Horticulture 2020 (updated Jun 2023). https://agritech.tnau.ac.in/crop_protection/tomato_diseases_11.html

[LM1] Utah State University Extension: Tomato leaf mold (Passalora fulva). https://extension.usu.edu/vegetableguide/tomato-pepper-eggplant/leaf-mold

[LM2] Ohio State University (High tunnel disease facts): Tomato diseases - leaf mold. https://u.osu.edu/hightunneldiseasefacts/?p=146

[LM3] Cornell AgriTech / Cornell Vegetables: Tomato leaf mold. https://vegetables.cornell.edu/crops/tomatoes/tomato-leaf-mold

[LM4] UF/IFAS Extension PP368: Common tomato high-tunnel production diseases in Florida (leaf mold caused by Passalora fulva and Pseudocercospora fuligena). https://edis.ifas.ufl.edu/publication/PP368/pdf

[MV1] Biological and molecular characterization of a new isolate of Tomato mottle mosaic virus causing severe shoestring and fruit deformities in tomato plants in India. Plants 2024, 13(19): 2811 (ICAR-NRRI, ICAR-IISR). https://www.mdpi.com/2223-7747/13/19/2811

[MV2] Tomato mosaic virus: transmission studies (Uttar Pradesh, India). Journal of Pharmacognosy and Phytochemistry 6(6) (2017). https://www.phytojournal.com/archives/2017/vol6issue6/PartG/6-5-135-757.pdf

[MV3] Drishti IAS (news explainer, secondary source): CMV and ToMV viruses in tomato crop in Maharashtra and Karnataka. https://www.drishtiias.com/pdf/1688722698.pdf

[MV4] EPPO Global Database: Tomato mottle mosaic virus - distribution in India (present, no details; intercepted on seeds from India). https://gd.eppo.int/taxon/TOMMV0/distribution/IN

[MV5] EPPO: Pest risk analysis quick scan for Tomato mottle mosaic virus (notes that ToMMV has been shown to break ToMV resistance). https://pra.eppo.int/getfile/531404a5-dcde-4d23-a2d6-cd09f49f0b5a

Items checked but not included

- MPKV Rahuri / PDKV Akola disease spray schedules for tomato: none were retrieved, so none are reproduced. MPKV material used here covers breeding objectives and the variety Phule Raja only.
- CIB&RC fungicides were cross-checked against the compilation up to 30/09/2025 (copy on a university website); CIB&RC insecticides were checked only against older versions (2019 / 2021), so vector-insecticide entries should be re-verified on the live register.
- Powdery mildew, buckeye rot and nematodes: not covered in this twelve-disease set (CIB&RC lists a buckeye rot claim for tomato).
- Registration changes after 30/09/2025 were not checked.